

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

Roll No. :S046	Name:Abhishek Yadav
Class :SY BSC IT	Batch : 01
Date of Assignment :22/07/2026	Date/Time of Submission :22/07/2026

Questions:

2a. Write a python code to create nested tuples.

Code:

```
*practical 2A.py - C:/Users/itlab/Desktop/Python for data structure/practical 2A.py (3.14.6)*
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student = ("Abhishek", 46, ("BSC IT", "Semester III"), 100)

print("Name :",student[0])
print("Roll no:",student[1])
print("Department :",student[2][0])
print("Semester :",student[2][1])
print("Percentage :",student[-1])

print("\nAbhishek_46")
```

Output:

```
Name : Abhishek
Roll no: 46
Department : BSC IT
Semester : Semester III
Percentage : 100

Abhishek_46
```

2b. Write a python code to sort the nested tuple using sorted () function.

Code:

```
tup = (1,3) , (4,5) , (2,1)
sorted_data = sorted(tup, key=lambda x: x[1])
print("sorted data:")
print(sorted_data)

print("\nAbhishek_46")
```

Output:

```
sorted data:
[(2, 1), (1, 3), (4, 5)]

Abhishek_46
```

2c. Write a python code to copy or clone list.

Code:

```
tup1 = ((1,3), (2,5), (6,2), (5,3))
tup2 = ()
tup2 = tup1
print("Original List:", tup1)
print("Cloned List:", tup2)

print("\nAbhishek_46")
```

Output:

```
Original List: ((1, 3), (2, 5), (6, 2), (5, 3))
Cloned List: ((1, 3), (2, 5), (6, 2), (5, 3))

Abhishek_46
```

2d. Write a python code to check immutability property of python tuples.

Code:

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```
tup1 = ((1,3), (2,5), (4,2), (5,3))
```

```
tup1[1] = (4,3)
```

```
print(tup2)
```

```
print("\nAbhishek_46")
```

Output:

Traceback (most recent call last):

File "C:/Users/itlab/Desktop/Python for data structure/practical 2A.py", line 3, in <module>

tup1[1] = (4,3)

TypeError: 'tuple' object does not support item assignment

